

RUMKEY CHOWDHURY

Assistant Urban Planner, Sheltech

Graduate Student, Urban and Regional Planning

Bangladesh University of Engineering and Technology

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ACADEMIC CREDENTIALS

- **Bachelor in Urban and Regional Planning | 2020 - 2025**
Bangladesh University of Engineering and Technology, Bangladesh
GPA- 3.84 out of 4.00
- **Higher Secondary Certificate (HSC) | 2017-2019**
Holy Cross College, Bangladesh | GPA- 5.00 out of 5.00

RESEARCH INTERESTS

- Transportation and Mobility
- Environmental Sustainability
- Disaster Management and Climate Change
- Smart Cities

RESEARCH EXPERIENCE

Influence of Built Environment on Electricity Consumption in Dhaka City

Nov 2023 – March 2025

Undergraduate Thesis

Supervisor: Dr. Mohammad Shakil Akhter (Professor, Dept. of URP, BUET)

Recently Worked:

- 1) Analysing electricity consumption trends for different usages across different income zones and spatial units in Dhaka.
- 2) Developing regression models using statistical techniques to quantify the impact of the built environment and spatial factors on electricity consumption.

PROFESSIONAL EXPERIENCE

- **Sheltech, Dhaka, Bangladesh** April 2025 – Present
Assistant Urban Planner of Improving Urban Governance and Infrastructure Project (IUGIP)
 - Data analysis and report preparation
 - Digitisation, geo-referencing and satellite image processing
 - Plan preparation and policy development
- **Local Government Engineering Department, (Intern), Dhaka, Bangladesh** July, 2024
I had the opportunity to learn about several key lessons:
 - ToR (Terms of Reference)
 - Inception report
 - Reviewing survey reports, survey maps and GIS data
 - Reviewing of upazila and municipality master plans
- **Sheltech and CEGIS, (Intern), Dhaka, Bangladesh**
I had the opportunity to visit the consultancy firms and learn about –
 - Drone survey and process
 - Image processing, analysis, and preparation of different categorised maps

ACADEMIC PROJECTS

1. **Identifying Hotspots and Severity of Road Accidents along National Highway (N1) and Analyzing the Influence of Corresponding Land Cover:** This study analyzes accident hotspots and severity along National Highway 1 (N1) using GIS-based kernel density estimation, creating land cover maps, examining changes over time, and ultimately observing correlations between land cover trends and accident severity.
2. **Application of Game Theory in Urban Planning Field using Python:** Applies game theory to urban planning using Python, focusing on strategic interactions in competitive and cooperative scenarios to inform decisions in areas such as transportation, land use, and resource management
3. **Traffic Impact Assessment of Central Hospital in Dhaka and Baridhara along Pragati Sarani Road:** Conducted TIAs for Central Hospital and Baridhara along Pragati Sarani, analyzing land use, road layout, traffic volumes, road capacity, and trip attraction. Used VISSIM simulation to assess current and future traffic scenarios, recommending strategies to improve access and support urban planning.
4. **Trip and Mode Choice Decision Making in Selected Routes within Dhaka:** Analyzed trip and mode choice decisions on selected Dhaka routes, focusing on travel time, costs, mode availability, and user experiences.
5. **Integrated Redevelopment and Block-Based Urban Design for DNCC Ward 21:** One study proposes a comprehensive redevelopment plan for DNCC Ward 21, emphasizing mixed-use and improved housing, healthcare, education, and commercial facilities while prioritizing mobility and accessibility. The other focuses on block-based urban design to optimize land use, increase open spaces, and promote walkability.
6. **Harmony in Green: A Comprehensive Regional Plan for Sustainable Forest Conservation and Socio-Economic Development:** Outlines a regional plan for sustainable forest conservation and socio-economic development in Bagerhat, Khulna, and Satkhira, focusing on agroforestry, eco-tourism, and community participation while protecting the environment with livelihood opportunities aligned with SDG 15.
7. **A Study on the Government, NGO, and CBO Initiatives for Rural Development in Bangladesh:** Explores rural development through government, NGO, and CBO initiatives, using KII, FGD, and surveys to assess impacts and community perceptions, highlighting key challenges and recommendations.
8. **Site Planning of Dhanmondi Lake Integrated Park and Neighbourhood Plan for Residential Area:** One plan focuses on addressing challenges in the park, offering recommendations to improve its design, facilities, and environmental management. The other plan for the Annaya residential area integrates housing units, roads, community facilities, open spaces, and commercial zones, adhering to standards.

AWARDS

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| • Awarded by “Dean’s list”, “University Merit” and “University Stipend” for three years | Jan 2020-July 2022 |
| • Higher Secondary School Certificate Board Scholarship | December 2019 |
| • Champion of Physics Olympiad (Junior) in Dhaka Residential Model College in Science Fest | September 2016 |

SKILLS

- **Programming Languages:** C, Python, R
- **Software Tools:** Microsoft Office packages, OriginPro, ArcGIS, AutoCAD, VISSIM, SPSS, Sketchup, Microsoft Project Professional, Adobe Illustrator, Machine Learning
- **Survey Skills:** Social and Physical survey, Engineering survey, Land use survey, Transportation survey

LEADERSHIP AND PRESENTATION SKILLS

- **Design Head** of IEEE BUET Student Branch
- **Vice-Chairperson** of IEEE Women in Engineering, BUET
- International Youth Summit Deputy Country Coordinator at Workplace Skills Development Academy, New Zealand

LANGUAGE SKILL

- **IELTS:** Overall Band Score 7 (**Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 6.5**)
- **Languages:** Bengali (Native), English (Advanced Level - C1), Hindi (Limited working proficiency)

CERTIFICATIONS

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| • The City and You: Find Your Best Place, Coursera, offered by the University of Toronto | June 2020 |
| • Building Custom Regional Reports with Google Analytics, Coursera Project Network | May 2020 |

REFERENCES

- **Dr. Mohammad Shakil Akhter**
Professor, Department of Urban and Regional Planning,
Bangladesh University of Engineering and Technology
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- **Dr. Ishrat Islam**
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